

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed, developed, and tested PLC programming for automated manufacturing systems, integrating **Rockwell and Siemens S7 controllers** to reduce manual operation errors by **18%**.
- Troubleshoot and repaired complex automation issues on live production equipment, diagnosing faults in **servo drives and vision systems**, restoring operational status within **95% of SLA targets**.
- Managed the full lifecycle of capital projects for controls automation, from justification and budgeting to design and installation, ensuring adherence to **Machinery Directive** and **CE compliance**, delivering projects **10% under budget**.
- Developed and implemented new monitoring tools for process improvement, defining and selecting appropriate **automation hardware and network protocols** (e.g., **EtherNet IP, Modbus**) which increased machine uptime by **12%**.
- Coached junior control engineers and technicians on troubleshooting techniques and continuous improvement efforts, enhancing team problem-solving capabilities and contributing to a **15% reduction in recurring technical issues**.
- Ensured electrical control panels were built to **EN60204-1 standards** and maintained critical spares lists for site automation hardware, guaranteeing **99% availability** of essential components.

SKILLS

Automation & Control Systems – PLC Programming (Rockwell, Siemens S7, Telemecanique), Robotics, Vision Systems (Cognex), Servo Drives, Motion Control, SCADA Systems, GAMP Documentation (URS, FDS, FAT, SAT)

Networking & Protocols – EtherNet IP, DeviceNet, ControlNet, Modbus, Profibus

Manufacturing Execution Systems – MES Integration, Plant IT Systems

Project Management & Compliance – Capital Project Management, Budget Management, EU Directives (Machinery Directive), CE Compliance, EN60204-1 Standards, GMP Environment, EHS Incident Investigation, Root Cause Analysis (RCA)

Software Engineering & Support – Java Spring Boot, Python, SQL (PostgreSQL, MySQL, Oracle), REST APIs, AWS, Docker, Linux, Git, Jenkins, Splunk, Dynatrace

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, R, Statistical Analysis

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Smart Factory Automation & Control Platform

- Designed and implemented a new automation control platform for a simulated pharmaceutical manufacturing line, integrating **Rockwell PLCs** and **Siemens S7 controllers** with high-speed robotics and **Cognex vision systems**.
- Developed fault-tolerant PLC code and robust error handling routines, achieving a **99.8% success rate** for automated assembly tasks and reducing critical production faults by **25%**.
- Configured and deployed **EtherNet/IP and Profibus** network infrastructure to support real-time data flow between PLCs, servo drives, and a central MES, enabling predictive maintenance alerts and improving overall equipment effectiveness (OEE) by **15%**.
- Managed project documentation including URS, FDS, and FAT/SAT reports, ensuring compliance with **GAMP guidelines** and internal quality standards for the pharmaceutical industry.

Financial Services Transaction Monitoring Dashboard

- Developed and maintained a **payments operations dashboard** using **Java Spring Boot** and **Python**, providing real-time transaction-flow monitoring for a financial services client.
- Implemented operational alerting for failed and stuck payment messages, reducing average incident response time by **30%** and ensuring adherence to strict Service Level Agreements (SLAs).
- Supported production environments with a focus on audit trails and **GDPR** awareness, collaborating with business analysts to define and triage operational requirements.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)